V93000 Specifications

Link Scale USB3 card

Preliminary Specifications

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Link Scale USB specifications

Use the Link Scale USB3 card to execute software-based functional tests and high-speed input/output scan tests on Smart Scale™ and EXA Scale™ test systems.

Scope of specifications

Note	The specifications are subject to change without notice. Make sure to refer always to the latest revision found in the most recent SmarTest documentation (System Reference > Specifications).

This specification covers the Link Scale USB3 card (part number E8018USB for Smart Scale™, EK806USB for EXA Scale™).

Advantest specifies and verifies the specifications at the DUT interface pogo level and verifies them during the performance verification before shipment. Appropriate DUT board design and layout including proper usage of the GND sense inputs and fixture delay measurements (TDR) is required to ensure specified performance at the device under test.

Specifications define the warranted technical performance covered by products within the stated adjustment interval and operating conditions. Typical specifications define the technical performance met by at least 80 % of all products during product qualification. A typical specification is not covered by product warranty and does not include measurement uncertainties.

Link Scale USB3 card overview

Link Scale USB3 is a digital card for the V93000 Smart Scale™ and EXA Scale™ generation of SoC test systems. It supports the following use cases as part of engineering or production test programs:

- Download, execution and tracing of software-based functional test content in binary (executable) form
- Scan test with data delivery and result upload through a high-speed input/output (HSIO) USB3.2 interface
- Basic DC test (continuity, impedance) through attached external resources.

Link Scale USB3 card

The Link Scale USB3 card uses a computer-on-module (COM) architecture to provide industry standard interfaces (USB3.2, JTAG, SWD) with full protocol stack support. Each COM runs an operating system with support for user-specific drivers and executables. The COMs interface with the system controller through a LAN connection and, if needed, trigger signals to/from other channels in the test system.

One Link Scale USB3 card contains four COMs, with each COM supporting one or two sites, depending on the operating system, device drivers and application software installed. The card provides a total of 16 ports connecting to the DUT through pogo pins, with each COM providing 2 high-speed (USB3.2) and 2 low-speed AUX ports. In addition, 2 UART outputs are available per COM.

As the card has no built-in PMU resources, a DC switch matrix allows to use external resources, like a per-pin PMU of a PS1600, for basic DC measurements like contact test.

Further information

For more information about the Advantest V93000 SOC Series, please visit *www.advantest.com*.

Contact information

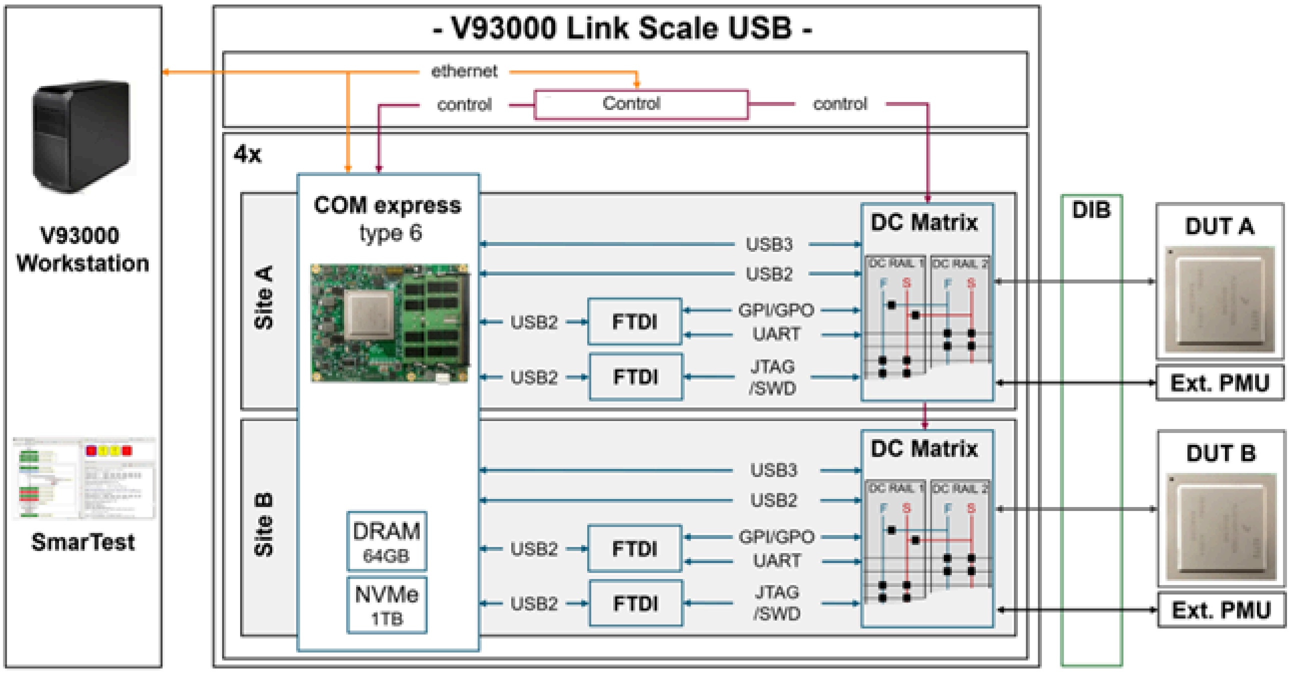
For more information about the Advantest V93000 SOC cards, please contact your local Advantest sales representative.

This information is subject to change without notice.

Link Scale USB signal assignment for one COM

Overview

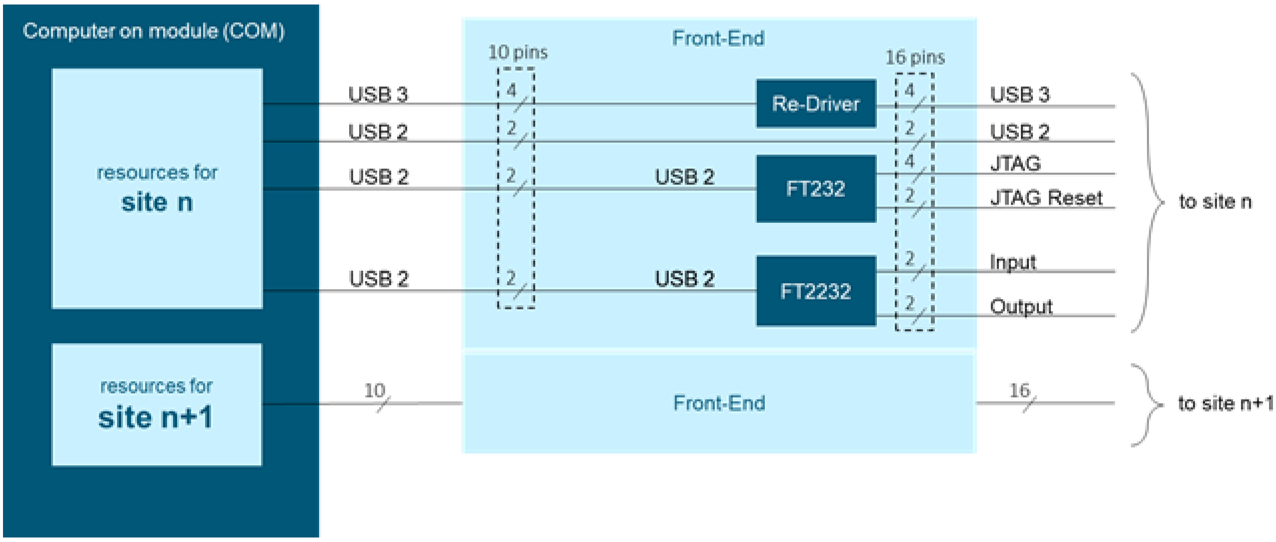
The figure below shows an overview of the signals for one COM. Four COMs are available per card.



Link Scale USB signal assignment for one COM

Signal assignment for a front-end

The figure refers to the two sites n and n+1, mapped to one COM.



Signal assignment for a front-end

Signal assignment for a pogo block

A Link Scale USB3 card connects to a DUT through four high-density (HD with 32 pins) pogo blocks. Refer to "Link Scale USB3 pogo pin assignment" (topic353563).

Each pogo block provides the signals for two sites. The table below refers to the two sites 1 and 2, mapped to one COM.

Signal	Pogo pin for site 1	Pogo pin for site 2
USB_SS_TX_p	A1	A9
USB_SS_TX_n	A3	A11
USB_SS_RX_p	A2	A10
USB_SS_RX_n	A4	A12
UART_TXD	A5	A13
GPI	A7	A15
UART_RXD	A6	A14
GPO	A8	A16
JTAG_TRST	B1	B9
JTAG_SRST	B3	B11
USB_p	B2	B10
USB_n	B4	B12
JTAG_TDI	B5	B13
JTAG_TMS	B7	B15
JTAG_TDO	B6	B14
JTAG_TCK	B8	B16

Link Scale USB basic parameters

Ports

Ports	
Total pin count	128 pins per card
USB 3.2 Gen2 lanes	8 lanes (2 per COM) per card, as 8 differential Rx and 8 differential Tx pairs
USB 2.0 lanes	8 lanes (2 per COM) per card, differential Rx/Tx combined
AUX ports (JTAG / SWD)	8 ports per card (2 per COM) with 5 pins each
UART ports	8 pins per card (2 per COM)
GPIOs	8 pins per card (2 per COM)

Speed

Speed	
USB 3.2 Gen2 ports	10.0 Gbit/s (USB 3.2) 480 MBit/s (USB 2.0)
AUX ports (JTAG / SWD)	15 Mbit/s
UART ports	12 Mbit/s
GPIOs	1 Mbit/s

Levels

Levels	
USB 3.2 Gen2 Ports	USB 3.2 compliant
AUX ports (JTAG / SWD)	1.2V 1.5V 1.8V 2.5V 3.3V
UART ports	1.2V 1.5V 1.8V 2.5V 3.3V
GPIOs	1.2V 1.5V 1.8V 2.5V 3.3V

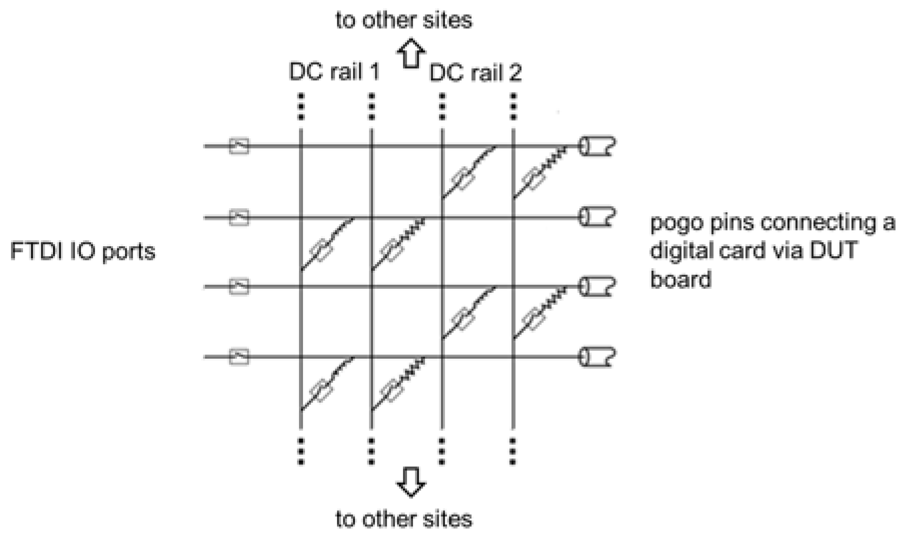
Operation and maintenance

For the definition of week, month, year in the context of a maintenance period refer to *Units of maintenance periods*.

Environment, operation, and maintenance	
Warm-up time	60 min
External adjustment period	6 months
Internal adjustment period, valid at ambient temperature within ±2.5 K of internal adjustment (auto calibration) temperature	3 months
Operating temperature range	+15 # to +30 # (+59 # to +86 #)
Temperature range for warranted specifications	+20 # to +30 # (+68 # to +86 #)
Maximum humidity at +30 # (+86 #)	< 70% relative humidity, non-condensing
Storage temperature range, without the presence of water	-40 # to +70 #C (-40 # to +158 #)

Link Scale USB DC characteristics

The Link Scale USB3 card does not contain built-in resources for DC measurements. Instead, it uses an internal switch matrix between the UART / AUX IO signals and the other ports to connect external resources, that is digital card channels, for DC measurements, such as continuity or (differential) impedance.



DC switch matrix of front-end

DC access can be done single-ended, or in a higher accuracy force/sense configuration (two channels).

Four DC rails per site are available as two force/sense pairs and can be used as follows:

- Two parallel one-terminal measurements per site, that is continuity
- Two parallel two-terminal measurements per site, that is leakage, with sensing
- One four-terminal measurement per site, that is differential impedance, with sensing.

Accordingly, up to four external channels (or two force/sense pairs) are needed per site for DC measurements. Hence, up to 32 external 'DC' channels are needed per card to support this.

The signal path through the Link Scale card can reduce the accuracy of the DC measurements by the following terms maximum.

Error terms	
Voltage measure	10 mV
Current force/voltage measure	10 mV + forced current × 25 Ω
Leakage current	20 nA

The inaccuracy introduced in one-terminal measurements due to the forced current can be minimized by using two-terminal measurements with dedicated terminals for each forcing and sensing.

Link Scale USB module PC (COM)

Module PC	
CPU	Intel Core i7-8665UE quad-core
DRAM	64 GByte (2 × 32 GByte Dual Channel DDR4-2400)
Mass Storage	950 GByte NVME